

CSCC43 Project:

Database Design Project

MyTix: User Manual

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1. Introduction

MyTix is a terminal-based ticket management system developed for event organizers and customers. The application supports user management, ticket purchasing, resale, event management, search functionality, reporting, and organizer decision support through an organizer pricing toolkit. Users interact with the system through a menu-driven command-line interface.

2. Getting Started

2.1. Prerequisites

- A Linux environment.
- MySQL installed and running.
- Java installed.
- All data project files (run.sh and .txt files) should use LF line endings. A .gitattribute file was added to ensure this.
- Make sure your local SQL configuration is setup in the following way:
 - URL= "jdbc:mysql://localhost:3306/mytix";
 - USERNAME = "root";
 - PASSWORD = "";
- If your local SQL configuration is NOT setup in the above manner, then update src/Database.java to match your local configuration

2.2. Compiling and Running

How to compile and run the program

On Linux: (run the following commands in order)

- `mysql -u root mytix < sql/drop.sql` // dropping schema
- `mysql -u root mytix < sql/schema.sql` // creating schema
- `sudo mysql --local-infile=1 mytix < sql/load.sql` // loading files
 - If error here: then you need to run this command in mysql terminal:
 - `SET GLOBAL local_infile = 1;`
 - `exit;`
 - Now run 2nd bullet point again
- `sh run.sh`

On Windows: (run the following commands in order)

- `Get-Content .\sql\drop.sql | mysql -u root mytix` // dropping schema (aka reset db)
- `Get-Content .\sql\schema.sql | mysql -u root mytix` //creating schema
- `Get-Content .\sql\load.sql | mysql --local-infile=1 -u root mytix` //loading files
 - If error here: then you need to run this command in mysql terminal:

- SET GLOBAL local_infile = 1;
 - exit;
- Now run 2nd bullet point again
- javac -cp "mysql-connector-java-8.0.29.jar" src*.java
- java -cp "src;mysql-connector-java-8.0.29.jar" Main

3. User Guide

3.1. Main Menu

The Main Menu is the entry point of the application. Users can navigate to User Management, Organizer Operations, Customer Operations, Searches & Queries, Reports, or exit the application by entering the corresponding menu number.

3.2. User Management

- Create User: Creates a new customer or organizer account by entering the user's email, name, address, date of birth, and account type
- Delete User: Deletes an existing user by entering their email address. Associated records are automatically removed through cascading deletes.

3.3. Organizer Operations

- Create Event: Creates a new event by entering the organizer's email, event name, and resale cap
- Add Performance: Adds a new performance by specifying the event, venue, date, and time
- Define Price Tier: Created a price tier for a selected performance.
- Assign Sections to Tier: Assigns venue sections to an existing price tier.
- Update Tier Price: Updates the price of a tier for a future performance when permitted.
- Block Seat: Blocks an available reserved seat for a performance.
- Unblock Seat: Removes a previously blocked seat.
- Cancel Performance: Cancels a performance and processes the associated ticket refunds.
- Organizer Pricing Toolkit: Suggests price tiers and a tier structure based on similar historical performances

3.4. Customer Operations

- Add Payment Information: Stores payment information for future purchases
- Purchase Tickets: Purchases one or more tickets for a selected performance
- Cancel Ticket: Cancels an eligible ticket and processes the refund.
- Create Resale Listing: Lists an owned ticket for resale.

- Withdraw Listing: Removes an active resale listing.
- Buy Resale Ticket: Purchases an available resale ticket.
- Add Review: Submits a review for an attended performance.
- View Ownership History: Displays the ownership history for a selected ticket

3.5. Searches & Queries

- Query 1: Search Performances By Distance - User enters Latitude, Longitude, a distance (Default = 15), and an option to rank the results by cheapest ticket (ASC, DESC) or distance from the search location. The query returns all upcoming performances scheduled at venues that are located within the specified distance along with performance, venue, ticket and distance information. (To compute distance, we used the Haversine formula to successfully calculate distance between two points).
- Query 2: Search Performances By Postal Code - User enters a postal code (5,6 characters) and the query returns all the upcoming performances at venues in the same and adjacent postal codes (for adjacent postal codes, the first 3 characters should match).
- Query 3: Search Performances by address - User enters a street address, city, country, and postal code. The query then returns the venue matching the entered address (if it exists), along with all upcoming performances at that venue.
- Query 4: Search by Date and Ticket Availability - User is asked a start date, end date, and the number of tickets they need. Query returns all the upcoming performances (along with venue information) within the date range that has at least the requested number of tickets available.
- Query 5: Advanced Filters Supported - Users can narrow down their search by applying multiple filters in any combination. Filters such as (city, segment, performance start and end date, ticket price range, number of tickets, type of section) are supported. If a user doesn't want to apply a filter they can just simply press **enter**. (Design Choice: Returns always upcoming performances - not the past ones).
- Query 6: Seat Map Summary: - User is asked for a performance ID and given if the performance ID exists, for every section, the results show its tier, price, the number of tickets sold or blocked, and the number of available seats / remaining capacity.
- Query 7: Find Best Available Seats - User enters a performance ID, the desired number of consecutive tickets, and optionally a maximum budget. The query returns the group of consecutive available seats in the same row with the lowest total price that satisfied the requested quantity and budget. If no such group exists, the user is informed that no suitable seats are available.

3.6. Reports

Generates the administrative reports listed in the project description. Refer to project guidelines for more details on each report. Some reports prompt the user for additional information such as a month, performance ID, or date range if applicable.

- Report 1: Total Tickets Sold & Gross Revenue: This report returns total number of tickets sold and the gross revenue in a specific date range (start and end dates should be entered) by either a city or by a venue within a city (Which is more specific and shows venue names alongside city, total tickets sold and gross revenue generated).
- Report 2: Total Events & Performances: Runs a report that returns a total number of events and performances 1) Per Segment and Genre, 2) Per Country, 3) Per Country and City and 4) Per Country, City and Venue.
- Report 3: Rank Organizers by Gross Revenue: This report contains three sub-reports with options to 1) Rank the organizers by their gross revenue overall, 2) Rank the organizers by their gross revenue per country and 3) Rank the organizers by their gross revenue per city (which includes each city).
- Report 4: Potential Ticket Scalpers: Identifies customers who, within the past year, listed more than half of the tickets they purchased for resale, provided they purchased at least ten tickets. The report displays the customer's city, email, number of tickets purchased, and number of tickets listed for resale.
- Report 5: Rank Customers By Orders: Customers can be ranked by the number of orders they placed in the specific time period (by entering start and end dates) or they can be ranked by the total number of orders they placed per city within the past year (given they placed at least 2 orders).
- Report 6: Cancelled Tickets & Performances Report: Returns customers with the largest number of cancelled Tickets and the organizers with the largest number of cancelled performances.
- Report 7: Sell-Through Reports: Provides three sell-through analyses. The first reports the sell-through rate for every performance, the second reports the sell-through rate for each price tier of a selected performance, and the third identifies performances within a specified month that either sold out or sold less than 25% of their sellable capacity
- Report 8: Resale Reports: Provides two resale analysis. The first reports the number of completed resales for every event, the average resale markup over the original face value, and the fraction of resale listings priced at the resale cap. The second reports the top ten events by resale volume within a user-specified date range.
- Report 9: Popular Review Phrases: Analyzes customer review comments and reports the most frequently occurring phrases for each event.

Report 1: Total Tickets Sold & Gross Revenue

2. Report 2: Total Events & Performances

3. Report 3: Rank Organizers by Gross Revenue

4. Report 4: Potential Ticket Scalpers

5. Report 5: Rank Customers by Orders

6. Report 6: Cancelled Tickets & Performances Report

4. System Limitations

- The application uses a terminal interface only, which provides limited usability compared to a graphical user interface.
- Input validation is minimal. Most of the queries and operations rely on the expectation that users enter valid data format, otherwise java throws an exception (e.g invalid date entered).
- Many operations require users to enter database IDs (e.g., performance IDs or venue IDs) rather than selecting from a list of available options or asking for names which are more typically known to users.
- The system doesn't implement user authentication or role-based access control. Any user with access to the application can perform customer, organizer, query, and reporting operations
- The Organizer Toolkit uses historical averages rather than predictive analytics for its recommendations.
- Report 9 uses a simple phrase extraction approach rather than true natural language processing (NLP) technique for noun phrase extraction.
- Error handling is primarily intended to prevent database inconsistencies rather than providing user-friendly feedback.

5. Possibilities for Improvement

- Develop a graphical or web-based interface to make navigation easier and more user-friendly.
- Add stronger input validation and clearer error messages for invalid dates, IDs, prices, and other user inputs.
- Implement authentication and role-based access so customers and organizers can only access the operations available to their account type.
- Improve the Organizer Pricing Toolkit using demand forecasting, seasonal trends, historical sales, and machine-learning techniques.
- Replace the basic phrase extraction used in Report 9 with a proper natural language processing method for identifying noun phrases.
- Add visual analytics dashboards with charts for sales, attendance, sell-through rates, resale activity, and customer trends.